

To Bot or Not? Conversational Agents and Authentic Heritage Experiences

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Abstract

Generative AI is rapidly reshaping heritage interpretation, with many museums adopting AI in the name of enhanced visitor engagement. Sector discourse, while championing responsible AI adoption, often celebrates AI's promises of interactivity, personalisation, and immersion. This paper disrupts that optimism through findings from a UK-based survey examining audience perceptions of conversational agents, a promising generative AI application in museums. While these agents can support engagement, our results reveal a complicated reality: 48% of participants voiced concerns about their use in museums. Our analysis identifies three core risks that challenge assumptions about AI's role in enhancing heritage engagement: its potential to distract from heritage collections, erode humanistic forms of engagement, and de-personalise visitor experiences. This paper provokes critical dialogue on how AI systems can be designed to sustain, rather than compromise authentic museum experiences, and in doing so, contributes to the sector's mission in modeling responsible AI adoption and use.

Keywords

Authenticity, Museums, AI, Conversational agents, Chatbots

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1 Introduction

AI is being increasingly discussed and adopted across the heritage sector: according to a Heritage Pulse Survey, as of September 2025, half of 115 UK heritage organisations surveyed have been experimenting with AI [5]. While AI is primarily being used by these organisations for operations management and heritage collections research, it is also being adopted specifically to support and enhance visitor engagement [5]. This includes using AI to produce more accessible content, enable interactions through conversational tools, and analyse visitor feedback [27]. Although AI is being heralded for transforming visitors' heritage experiences in relation to interactivity, immersion and personalization (e.g., [8, 9, 24, 25]),

discussion has also been rife with ethical concerns having to do with bias, mis/dis information, copyright, and labour [27]. These concerns are reflected in heritage policies and discourse which place pressure on museums to model responsible AI adoption and use [23, 29]. Heritage policies about AI, champion notions of institutional transparency, human oversight, and inclusion – albeit very broadly across organizational types and AI systems. While such guidance remains high-level and the drive for innovating AI-mediated heritage experiences persists, idealistic notions regarding the perceived capacity of AI could overshadow responsible museum practices. As seen via the aforementioned Heritage Pulse survey, 'More than 60% of respondents now believe AI offers more opportunities than risks for their organisations – a jump of 20 percentage points since 2023' [6, para. 2].

In this paper, we complicate idealistic notions of AI for enhancing visitor engagement by demonstrating how conversational agents may prevent audiences from having 'authentic' heritage experiences. Authenticity has long been a topic of concern in museum contexts, which broadly speaking, is a term applied to something that is perceived to be 'real', or 'true' [11]. In heritage contexts, there have been diverse perspectives and diverging threads of discussion regarding what constitutes and affects authenticity. Despite diverse perspectives, authenticity is commonly positively associated with museum visitors' ability to make meaning, learn, cultivate self-identities, and to trust in museum organisations [28]. To challenge idealistic notions of AI for enhancing visitor engagement, this provocation summarizes theoretical perspectives on authenticity in heritage contexts and discusses findings from the authors' ongoing research.

Our research investigates the design of conversational agents for museum-based visitor engagement, starting with audiences' perceptions of conversational agents. These perceptions were collected as part of a survey distributed in the UK and surface opinions regarding how conversational agents may enhance but also hinder visitors' authentic museum experiences. Our analysis reveals three tensions that challenge assumptions about AI's role in heritage: its capacity to (1) distract from collections, (2) erode humanistic engagement, and (3) de-personalise experiences. We theorise the potential wider negative implications of these findings for heritage organizations and demonstrate how they may inform a sector-wide push for responsible AI adoption and use.



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2 Background Literature

2.1 From Authentic Objects to Authentic Experiences

In a global context within which AI is being used to generate synthetic content and mis- and dis- information, amplifying what was already considered to be an ‘environment characterized by pervasive deceit’ [22, para. 3], museums may be looked towards as pillars of reliable knowledge [3]. A poll in 2024 showed that the UK public still holds significant trust in museums – with 83% of respondents stating that they trust museum curators and even more so than other professionals such as judges and scientists [10]. As described in a UNESCO report ‘Recommendations on the ethics of AI’, trust in museums is founded on the credibility and integrity of the works they hold and the stories they tell [29].

To frame out study’s findings regarding the potential effects of AI on authenticity and in turn, trust in museum organisations, we must first trace (in brief) changes to the concept of authenticity in heritage spaces. Authenticity in museum contexts has traditionally been associated with material heritage objects that are ‘original’ and ‘real’ and therefore, have a unique ‘aura’ which is felt by visitors [2]. By aura, we refer to the sector’s privileging of material and ‘real’ objects due to their ability to provide visitors with ‘the visceral thrill of being in the presence of the original’ [1, p. 21]. This perception of authenticity as being rooted in a verified ‘original’ object has been referred to as ‘material authenticity’ [18]. However, authenticity even from this perspective was not only centred on the materiality of an original object – but also the associated information (scientific or irreputable facts) that have ‘accuracy’ and ‘coherence’ [7, p. 364]. Museum labels have always been key to conveying knowledge, wisdom, order, and meaning, and in turn, credibility [26, pp. 76-81], which has been described as ‘factual authenticity’ [18].

A focus on material authenticity in the heritage sector has traditionally overshadowed other intangible aspects of authenticity (e.g., emotion, spirituality, practices), having long-lasting effects on contemporary museum practices [16]. As explained by [15], international policies such as the Nara document which govern heritage preservation ‘retain(s) a strong emphasis on form, design, materials, substance, use, function, tradition, and technique, even though intangible heritage, spirit, and feeling have been added’ [15, p. 135]. The visitor-turn in museum studies (a shift in organisational focus from material collections to visitor-centred experiences), which was propelled by the increasing use of digital technologies [32], complicated notions of authenticity even further. Digital technologies for supporting heritage engagement challenged traditional notions of ‘authenticity’ as rooted in the materiality or ‘aura’ of an original physical museum object: Professionals began to debate if digital collections retained an ‘aura’ or an ‘authenticity’ raising critical questions regarding what was ‘genuine, trustworthy, reliable and valid about the museum experience’ [26, p. 64]. As such, authenticity, where it is located when materiality shifts to virtuality, and how it is then experienced, has become ambiguous. However, building on these shifts in museum studies, recent attention to affect in curation and visitor engagement has prompted new understandings of authenticity. Attention to affect has positioned the subjective meaning-making and emotional responses of visitors as central to the heritage experience [31].

As part of this ‘affective’ wave, it has been argued that museum displays with limited material or factual authenticity can still entice a feeling of authenticity, described as ‘emotional authenticity’ [18]. Emotional authenticity is described by scholars as connoting a personal connection to heritage or empathy [28], and an experience that is felt to be real, having ‘integrity’ and ‘trustworthiness’ [12, p. 1396]. Speculation and artifice have always been part of museum displays and have been used for inspiring emotion, understanding, or immersion – a feeling of something that is real or ‘true’ [18, 26].

As reflected on in their study of science museum narratives: ‘Models, reproductions, facsimiles, and actors all look like the “real thing” and thus help to generate the feeling of authenticity, even though they are not materially authentic (in terms of originality) or factually authentic (as they may be based on guesswork or artistic license)’ [18, p. 40]. From this brief background, we can now consider authentic museum experiences as a complex interaction between material, factual, and emotional authenticity, which can be affected by a range of digital and non-digital interpretation techniques.

2.2 Conversational Agents and Authentic Heritage Experiences

While AI has been idealized for enhancing visitor engagement, the above contextualization of authenticity suggests a tenuous relationship between technology and heritage. Arguments for the use of conversational agents in heritage contexts are rooted in a history of museum social and interactive education, reflected in ‘[a]ttempts to use chatbots as a means of encouraging closer connections to collections and enhancing social interaction’ [4, p. 251]. As recent research has shown, conversational agents can also be a more accessible or inclusive way for audiences to engage with heritage – chatbots can empower visitors to ask questions [24], support personalised learning about heritage collections and topics [8, 19, 21, 25] and to have emotional experiences [9, 20]. This social and interactive learning perspective suggests that there is ample opportunity for conversational agents to contribute to visitor engagement, including authentic heritage experiences by supporting an experience that is felt to have integrity and be ‘real’ or ‘true’.

However, the relationship between conversational agents and authentic heritage experiences is complex: An interaction with a conversational agent is intended to mimic a form of human-human interaction, symbolising in itself - an ‘artificial exchange’. On the surface, such an interaction would appear to be in opposition to an authentic heritage experience encompassing a sense of ‘realness’. In fact, this artificial simulacrum has been argued to be an unethical conversation or exchange contradictory to museum experiences, in part, because chatbots cannot experience empathy that would normally be conveyed by museum staff [4].

Additional concerns in relation to authenticity and conversational agents in heritage contexts have also been raised: conversational agents are a way to mediate collections through ‘deception’, which challenges institutional transparency regarding what is real or fake [4]. Questions have further been flagged regarding the authenticity of re-mediations and re-enactments through algorithmic

systems, or also similarly referred to as ‘after-lives’, and simultaneously how they pose contradictions to the museum as an organisation founded on truth, ethics, and responsibility [17]. Additional ethical concerns include whose voice is prioritised through conversational agents and the ownership of the deceased’s memory and its commercialisation [17]. While this highlights mainly the opportunities and hints at risks posed by professionals and researchers regarding conversational agents, there is a need for further insight into audiences’ willingness to engage with this technology and perceptions around ‘real’ and ‘true’ heritage experiences.

3 A Note on our Ongoing Research

Our research has been exploring the design of chatbots in heritage contexts for learning – starting with audiences’ perceptions and subsequently, those of heritage professionals. This paper draws on initial findings from an audience survey, approved by the University of Strathclyde’s ethics committee [ID: 2601]. A survey was chosen to reach a broad UK public with various experiences with heritage and conversational agents. The survey was distributed to UK residents through a recruitment platform - Prolific. The survey included a linked participant information sheet and required participants to consent to the terms of participation prior to beginning. The survey was opened for 20 prolific participants at a time and across a range of days and times between 5 - 11 September 2024 in order to gather a more diverse participant pool.

The survey consisted of four open-ended questions about the perceived value of chatbots for a museum experience, expected functions, concerns, and an optional textbox for participants to expand on why they would be interested or uninterested in using conversational agents in museums. A series of ranking questions about the design of chatbots was also asked: the differences between the options in the ranking questions were minimal, and as such, we report here on the qualitative responses and aim to expand on these in future research. The survey questions were framed by Huebner’s ‘Theoretical Model: Social Media as a Museum Education Tool’, which captures three dimensions of digital tools in museums related to learning [14]. The model prompts attention to the 1) Use of the tool: value, barriers, and concerns of users; 2) Presentation of the tool: user preferences for presentation of cultural heritage content; and 3) Design of the tool: how it can be interacted with. The qualitative responses were analysed using a thematic analysis following the steps of [30], revealing three core tensions or themes described below as jumping off points for catalysing further research.

The analysis was done via a back-and-forth grounding in the data and related literature through inductive qualitative analysis (open coding): this required remaining close to the data initially, then collating these codes into larger categories, and subsequently, themes that spoke to the research questions. In doing so, the researchers adhered to a back-and-forth process of analysing codes, consulting literature, and collating categories into the three main themes [30]. We collected 200 responses from participants living in the UK covering a variety of ages and genders: our participant sample had slightly more female participants (54%) compared to male participants (45%) and a few who identified as another gender (2%). The participants were mainly in the age ranges of 25-34 (37%),

followed by 35-44 (28%), and 45-54 (18%), with a smaller number of participants in the age ranges of 18-24 (8%), 55-64 (7%) and 65+ (3%). In order to understand participants’ experiences with museums and how this may shape their responses, we asked participants how frequently they visit museums: several participants (43%) visit several times a year, and the second most common frequency was once a year (34%). We similarly asked participants how frequently they use chatbots such as ChatGPT/Co-pilot: Several participants use them several times a year 72% while 12% of participants reported not using chatbots at all.

4 Preliminary Findings on Conversational Agents, Heritage and Authenticity

When investigating audiences’ perspectives, open-ended survey responses painted a picture of envisioned museum experiences and simultaneously how conversational agents may contribute to but also significantly take away from these expectations. When asked to identify the perceived value, describe their interest/uninterest in, expected functionalities, and perceived concerns regarding conversational agents in museums, three thematic potentials, but also concerns, emerged regarding the ability of conversational agents to: 1) distract from heritage collections, 2) take away from humanistic heritage experiences, and 3) de-personalise heritage experiences. These tensions speak to expectations associated with the emotional, material, and factual dimensions of an authentic museum experience. When asked if participants had concerns about using chatbots in museums, 103 participants (52%) had no concerns, while 48 had concerns (24%) and 49 may have had concerns (24%).

4.1 Distract from Heritage Collections

Survey participants expressed concern regarding how conversational agents may shape their in-person experience in museums (28% of participants who expressed concerns). Specifically, participants suggested that conversational agents may interrupt the museum as a space to digitally unplug, to be in the moment, and to connect to heritage on display. In relation to digitally disconnecting one participant explained: ‘it [a chatbot] would mean having to get your phone out and use this whilst trying to enjoy the exhibition’. Others voiced concerns regarding how they distract from the experience: ‘it [a chatbot] may take away some of the spectacle of seeing things for the first time’, or as suggested by another - ‘people might spend too much time on their phone as opposed to taking it in’. A perception of the museum experience as centred on collections and materiality aligns with a traditional notion of authenticity as rooted in original material collections that may connote a certain spectacle or ‘aura’. However, simultaneously, conversational agents were also perceived by participants as potentially contributing to new dynamic connections with collection objects.

4.2 Take away from Humanistic Heritage Experiences

Participants also expressed concerns regarding the impact of a conversational agent on humanistic experiences in museums (20% of participants who expressed concerns). This concern overlaps with notions of an emotional authenticity – or something that resonates as real, encompassing empathy and personal connections.

Participant responses reflected concerns about how conversational agents may impact the human perspective, soul, and narratives that characterise heritage: As one participant responded, 'I like to learn new information and chatbots supply this, however, I also love human opinion when it comes to art and sculptures and their interpretations of things which chatbots can't supply'. This limitation was also associated by some participants with concern for detracting from staff's expertise: 'I don't want it (chatbots) to take jobs from knowledgeable people...' On the other hand, participants expected conversational agents to be social/friendly and to support a humanistic interaction and social experience with heritage.

4.3 De-personalise Heritage Experiences

Participants primarily suggested concern about how conversational agents could de-personalise heritage experiences by not having quality information, and even conveying misinformation, and the balance of personalisation with their data use and privacy (57% of participants expressed concerns). This concern overlaps with factual authenticity or the ability of visitors to access information grounded in accurate and reliable facts. For example, one participant responded: 'Chatbots don't always answer your questions properly so they might not be able to answer specific questions you have about the exhibitions'. Similarly, another suggested it would hinder experiences if 'it didn't work properly or didn't know the answer to any of your questions'. According to some participants, their interest and willingness to engage with a conversational agent would be dependent on how their data is used and stored: 'I would have privacy concerns regarding personal information.' Another participant explained they would be interested in using a chatbot 'depend[ing] on how much personal info is required / confidentiality etc.' However, participants also suggested, despite this risk, that interactions with conversational agents could enable personalised experiences by fulfilling curiosity and enabling deep dives into information.

5 Future Research Directions and Initial Design Implications

As museums are increasingly adopting AI and are becoming more confident in their potential to contribute to visitor access and engagement, we call for further timely attention to the risks it poses for authentic heritage experiences. Several high-level policies place pressure on museums to adopt AI responsibly and enact principles such as transparency and human oversight, without clear relevance for day-to-day practices. In conjunction with this ambiguity and a drive for innovation, the careful practices of museums may be side-lined. Our study demonstrates the potential of conversational agents for supporting authentic experiences but highlights the lesser discussed risks, including its potential to 1. distract from heritage collections, 2. detract from a humanistic heritage experience, and to 3. de-personalise a museum visit.

The findings prompt us to consider how the design of conversational agents can align with audiences' expectations of a museum experience. Conversational agents could support rather than distract from dynamic connections with material collections, by supporting novel ways of engaging with objects (via multi-media) and by provoking attention to the 'spectacle' on display (a practice

referred to as indexing [13]). To mitigate de-personalisation, developers could strive for 'quality' information by using trusted and curated data from heritage organisations, restricting knowledge boundaries to relevant topics (an example is the London Museum's conversational search - Clio¹), and making the use of visitors' data transparent. Humanistic experiences, grounded in emotion that enhance rather than subvert human interaction and interpretation, could be supported, for example, by agents that emphasise personal narratives and elicit interactions between users and other visitors or friends/family. These concerns related to authenticity could inform not only future design but also benchmarks for evaluating AI systems, moving from high-level guidance to practical insights to sustain museums' responsible AI adoption and use.

The thematic tensions identified also expose that some visitors may still be grappling with digital transformations in museum spaces and what this means for meaning-making grounded in notions of virtuality, interaction, and emotion. However, these perceptions of conversational agents and their impacts on authentic experiences could be different across visitors from various cultural backgrounds. A follow-up study may explore perceptions of authenticity between diverse user groups interacting with conversational agents. If designs fail to support authentic heritage experiences, there is consideration for not only the effect this may have on visitor meaning-making and learning but also trust in heritage organisations. Therefore, we also call for further research into how dimensions of authenticity can be supported through the design of conversational AI, and what benchmarks of authenticity may be used to assess and evaluate AI-heritage systems.

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¹ More info about Clio can be found here: <https://www.londonmuseum.org.uk/blog/introducing-our-conversational-search-agent/>

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